

# EC400: Optimization & Fixed Points

## Exam 2020

1. (20 marks) For  $(a, b) \in \mathbb{R}^2$ , consider the following quadratic form

$$Q(\mathbf{x}) = a(x_1^2 + bx_2^2 + bx_3^2) + 2x_2x_3.$$

- (a) (3 marks) Identify the *symmetric matrix of coefficients*  $A$  associated with this quadratic form. That is, find the matrix  $A$  satisfying  $Q(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}$ . Then, find all the principal minors of  $A$ .
  - (b) (2 marks) Define what it means for  $A$  to be *indefinite*, *positive semidefinite* and *negative semidefinite*.
  - (c) (15 marks) Find the values of the parameter values,  $a$  and  $b$ , such that the quadratic form  $Q(\mathbf{x})$  is respectively *indefinite*, *positive semidefinite* and *negative semidefinite*? Then, plot these regions in  $\mathbb{R}^2$ .
2. (20 marks) Consider a function  $f : U \rightarrow \mathbb{R}$  for some convex set  $U$  in  $\mathbb{R}^n$ , and answer the following questions:
- (a) (14 marks) Prove that if  $f(\mathbf{x})$  is *quasi-concave*, then the function  $h(\mathbf{x}) = g(f(\mathbf{x}))$  is *quasi-convex* for any *strictly decreasing* transformation  $g : \mathbb{R} \rightarrow \mathbb{R}$ .
  - (b) (6 marks) Let  $U = \mathbb{R}_{++}^3$ , consider the function

$$h(\mathbf{x}) = -\exp(x_1^2 x_2^4 x_3^6).$$

Is the function *quasi-convex*?

3. (40 marks) For  $a > 0$ , consider the problem:

$$\max_{x,y} y^{1/2} x^{1/2} \quad \text{s.t.} \quad y + ax \leq 1/4, \quad y \geq 0, \quad x \geq a.$$

- (a) (6 marks) Is the objective function concave? Are the constraints convex? For what values of  $a > 0$  is Slater condition satisfied? So, for what values of  $a > 0$ , can you solve this optimization problem using the *Kuhn-Tucker method*?
- (b) (6 marks) Suppose the restrictions on  $a$  you found in first part of the problem hold. For  $y > 0$ , write down the *Kuhn-Tucker necessary first order conditions* that must be satisfied by the solution of the constrained optimization problem. Are solutions to these conditions also maximizers of the Lagrangian?
- (c) (20 marks) Suppose the restrictions on  $a$  you found in first part of the problem hold. Solve the constrained optimization problem in terms of  $a$ .
- (d) (8 marks) Suppose the restrictions on  $a$  you found in first part of the problem hold. Use the *Envelope Theorem* to find the slope of value function with respect to the parameter  $a$ , and discuss its sign.

4. (20 Marks) Consider the correspondence  $\mathbf{h} : [0, 1]^2 \rightrightarrows [0, 1]^2$ , where  $\mathbf{h}(\mathbf{x}) = (h_1(\mathbf{x}), h_2(\mathbf{x}))$  and

$$h_1(\mathbf{x}) = \begin{cases} 1 & \text{if } x_2 \in (a, 1] \\ [0, 1] & \text{if } x_2 = a \\ 0 & \text{if } x_2 \in [0, a) \end{cases} \quad \text{and} \quad h_2(\mathbf{x}) = \begin{cases} [1/2, 1] & \text{if } x_1 = 1 \\ 1/2 & \text{if } x_1 \in (a, 1) \\ [0, 1/2] & \text{if } x_1 = a \\ 0 & \text{if } x_1 \in [0, a) \end{cases},$$

for a scalar  $a \in (0, 1)$ . Briefly, argue why *Kakutani fixed point theorem* applies. Then, for all values of  $a \in (0, 1)$  find all the *fixed points* of this correspondence.